Fakultas Informatika
School of Computing
Telkom University

Aplikasi Berbasis Platform

Node.js

NodeJS

- Created in 2009
- JavaScript runtime environment that executes JavaScript code server-side, outside of a web browser
- Allows us to become a full-stack developer just by learning one programming language, JavaScript

JAVASCRIPT
JAVASCRIPT EVERYWHERE
makeameme.org

Node.js

Process Object

Node.js Global Object Member that contains properties and functions that **provide information about currently running processes**

Modularization

Technique of **separating code into modules** that are independent but can be used with each other to build a complex program

Events

Node.js applications are usually known to have an **event-driven pattern**. To create it, you need to know how to **create and raise an event** in Node.js

Filesystem

Back-end development will often **access or write files** on the computer. Using a filesystem, you can create feature such as upload files, writing logs.

Stream

Continuous processing of data as it is generated or received.

Node Package Manager (NPM)

Package manager for JavaScript and is the default package manager for Node.js

HTTP

HTTP module to **create HTTP server** in Node.js

Simple server with Node.js

```
// load http module
const http = require('http');

// create a HTTP server
const server = http.createServer(requestListener);

// run the HTTP server
server.listen(3000, 'localhost', () => {
   console.log('Server running on http://localhost:3000');
});
```

Routing with Node.js

```
const requestListener = (request, response) => {
   response.setHeader('Content-Type', 'application/json'); // set the content-type manually
   response.statusCode = 200; // set the status code manually

   const { method, url } = request;

   // Routing response based on url & HTTP verb/method manually using if-else
   if (url === '/' && method === 'GET') {
       response.end(JSON.stringify({ message: 'You are doing GET' }));
   } else if (url === '/' && method === 'POST') {
       response.end(JSON.stringify({ message: 'You are doing POST' }));
   } else {
       response.end(JSON.stringify({ message: 'Unknown action'}))
   }
};
```

Deploy web application in Node.js

- Run the terminal/console/command prompt
- Go to folder where you put the simple server code
- Run the command: *node your-server-code-filename*
  - If your filename is server.js, then

    ```
    node server.js
    ```
- Open browser with address below:
  - localhost:3000
  - or, 127.0.0.1:3000

Any question?

References

- https://nodejs.org

Fakultas Informatika
School of Computing
Telkom University

THANK YOU